

Correcting Forward Model Mismatch in Coded Aperture Snapshot Spectral Imaging via Two-Stage Differentiable Calibration

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Abstract. Forward model mismatch—sub-pixel mask misalignment and dispersion drift between the coded aperture and detector—is unavoidable in deployed CASSI systems, yet even moderate perturbations degrade state-of-the-art mask-guided transformers (MST) by over 16 dB. We present a self-supervised two-stage differentiable calibration pipeline that recovers 5-parameter mismatch from a single measurement and nominal mask alone, requiring no ground truth. Stage 1 performs a coarse hierarchical grid search scored by GPU-accelerated GAP-TV; Stage 2 applies gradient refinement through an unrolled differentiable forward operator using a Straight-Through Estimator (STE) for integer dispersion offsets. Evaluating five reconstruction methods across four scenarios on 10 KAIST scenes, we uncover a *mask-sensitivity spectrum*: mask-guided transformers (MST-S/L) suffer catastrophic degradation (>15 dB) but recover $\sim 48\%$ of the oracle gap ($\rho \approx 48\%$), deep prior methods (HDNet) show inherent robustness ($\rho \approx 0\%$), and iterative methods show intermediate sensitivity ($\rho \approx 13\text{--}23\%$). This spectrum provides a practical design guideline: deployed systems with limited calibration infrastructure should prefer robust reconstructors, while well-calibrated systems benefit most from mask-guided architectures. Code and results are publicly available.

Keywords: CASSI · Operator mismatch · Differentiable calibration
· Straight-Through Estimator · Hyperspectral imaging

1 Introduction

Deep learning has transformed hyperspectral image reconstruction, with mask-guided transformers (MST) [5,6] achieving >34 dB on the KAIST benchmark [8]. Yet these reconstructors depend on accurate knowledge of the forward operator—a dependency that creates a *sim-to-real gap* between the idealized model used during training and the physical system at deployment [13]. In coded aperture snapshot spectral imaging (CASSI) [14,1], this gap manifests as mask-detector misalignment and dispersion drift, degrading MST-L by over 16 dB (Figure 1).

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The CASSI forward model maps a hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H \times W \times A}$ to a 2D measurement $\mathbf{y} \in \mathbb{R}^{H \times W'}$:

$$y(i, j) = \sum_{k=1}^A m(i, j) \cdot x(i, j - d_k, k) + n(i, j), \quad (1)$$

where m is the coded aperture, $d_k = k \cdot s$ is the spectral dispersion offset for band k with stride s , and n is measurement noise. In practice, five parameters characterize the dominant misalignment: mask translation $(\Delta x, \Delta y)$, rotation θ , dispersion slope a_1 , and axis angle α .

The calibration challenge. Correcting this mismatch is difficult because: (1) spectral dispersion offsets d_k are integers, making the forward operator non-differentiable; (2) mask affine and dispersion parameters interact through the measurement model; and (3) in deployment, neither the true parameters nor the ground truth scene are available—calibration must be self-supervised from the measurement alone.

Contributions. We address these challenges with:

1. A **differentiable CASSI forward model** using a Straight-Through Estimator (STE) [3] for integer dispersion offsets, enabling end-to-end gradient-based calibration (Section 4).
2. A **two-stage self-supervised calibration pipeline** (Algorithm 1): coarse grid search followed by gradient refinement, recovering 5-parameter mismatch from a single measurement without ground truth.
3. A **mask-sensitivity spectrum** characterizing how five reconstructors respond to mismatch and calibration: mask-guided methods (MST-S/L) suffer >15 dB degradation but recover $\rho \approx 48\%$ of the oracle gap; deep prior methods show inherent robustness; iterative methods show intermediate sensitivity (Section 5).
4. A **four-scenario evaluation framework** (Ideal, Assumed, Corrected, Oracle) with complete open-source benchmark on 10 KAIST scenes across five methods.

2 Related Work

CASSI reconstruction. Classical approaches including GAP-TV [16,11] use alternating projection with total variation regularization. Plug-and-play methods replace the hand-crafted prior with learned denoisers [18,19], combining ADMM/GAP optimization with deep spectral denoisers. End-to-end deep networks have significantly advanced quality: HDNet [9] uses dual-domain unfolding, MST [5,6] introduces mask-guided spectral-wise attention, CST [4] applies coarse-to-fine sparse transformers, and DAUHST [7] combines degradation-aware unfolding with half-shuffle attention; see Yuan et al. [17] for a survey. All assume perfect forward operator knowledge.

Calibration in computational imaging. Arguello and Arce [2] optimize coded apertures for colored-CASSI but do not address post-fabrication misalignment. Traditional calibration requires external targets or careful laboratory procedures. Self-calibration from measurements alone has been explored for phase retrieval [12] and differentiable rendering [10], where end-to-end optimization through differentiable forward models has shown promise.

Differentiable forward models. Physics-based learned design [10] optimizes optical elements end-to-end but requires continuous relaxations. Deep unrolling [13] embeds the forward operator into network layers, making learned reconstructors sensitive to operator errors. HyperReconNet [15] jointly optimizes mask design and reconstruction but does not address post-fabrication calibration.

Our work is the first to combine an STE-based differentiable CASSI forward model with gradient-based self-supervised calibration targeting mask-detector misalignment in deployed systems.

3 Problem Formulation

3.1 CASSI Forward Model

The SD-CASSI (single-disperser) forward model maps a hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H \times W \times \Lambda}$ to a 2D measurement $\mathbf{y} \in \mathbb{R}^{H \times (W + (\Lambda - 1)s)}$:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}; \mathbf{m}, \{d_k\}) = \sum_{k=1}^{\Lambda} \text{shift}_{d_k}(\mathbf{m} \odot \mathbf{x}_k) + \mathbf{n}, \quad (2)$$

where $\mathbf{m} \in \{0, 1\}^{H \times W}$ is the coded aperture, $d_k = k \cdot s$ is the integer dispersion offset for band k , s is the stride (typically 2), and shift_{d_k} shifts the column index by d_k pixels.

3.2 Mismatch Parameterization

We model CASSI operator mismatch as a 5-parameter perturbation combining mask misalignment and dispersion drift:

$$\tilde{\mathbf{m}} = \mathcal{W}(\mathbf{m}; \Delta x, \Delta y, \theta), \quad \tilde{d}_k = a_1 \cdot k \cdot \cos \alpha, \quad \tilde{d}_k^y = a_1 \cdot k \cdot \sin \alpha, \quad (3)$$

where $\mathcal{W}$ applies bilinear-interpolated translation $(\Delta x, \Delta y)$ and rotation θ about the mask center, a_1 is the actual dispersion slope (nominal $s = 2.0$ px/band), and α is the dispersion axis angular offset. The true measurement uses the misaligned mask $\tilde{\mathbf{m}}$ with dispersion slope a_1 , while reconstruction assumes the nominal mask $\mathbf{m}$ with stride s .

3.3 Calibration Objective

Given measurement $\mathbf{y}$ (generated with unknown true parameters $\boldsymbol{\psi}^*$) and nominal mask $\mathbf{m}$, we seek:

$$\hat{\boldsymbol{\psi}} = \arg \min_{\boldsymbol{\psi}} \|\mathbf{y} - \mathcal{A}(\mathcal{R}(\mathbf{y}, \tilde{\mathbf{m}}); \tilde{\mathbf{m}}, \{d_k\})\|^2, \quad \tilde{\mathbf{m}} = \mathcal{W}(\mathbf{m}; \boldsymbol{\psi}), \quad (4)$$

where $\mathcal{R}(\mathbf{y}, \tilde{\mathbf{m}})$ is a reconstruction algorithm (GAP-TV in our pipeline) that produces a spectral cube estimate from measurement $\mathbf{y}$ using mask $\tilde{\mathbf{m}}$. This is self-supervised: minimizing the measurement residual requires no ground truth.

4 Method

4.1 Differentiable CASSI Forward Model

The key challenge is that dispersion offsets $d_k = k \cdot s$ are integers, making shift_{d_k} non-differentiable. We address this with a Straight-Through Estimator (STE) [3]:

$$\hat{d}_k = \text{round}(d_k), \quad \frac{\partial \hat{d}_k}{\partial d_k} \equiv 1. \quad (5)$$

In the forward pass, offsets are rounded to integers for exact indexing; in the backward pass, gradients flow through as if rounding were the identity function. This enables gradient-based optimization of parameters that influence the dispersion model.

The differentiable mask warping $\mathcal{W}(\mathbf{m}; \boldsymbol{\psi})$ uses PyTorch’s `affine_grid` and `grid_sample` with bilinear interpolation, providing exact gradients for Δx , Δy , and θ . The sign convention matches scipy exactly: $t_x = -2\Delta x/W$, $t_y = -2\Delta y/H$.

4.2 Differentiable GAP-TV Solver

We unroll K iterations of GAP-TV into a differentiable computation graph:

$$\mathbf{r}^{(t)} = \mathbf{y} - \mathcal{A}(\mathbf{x}^{(t)}; \tilde{\mathbf{m}}, \{d_k\}), \quad (6)$$

$$\mathbf{x}^{(t+1)} = \text{TV}_{\sigma}(\mathbf{x}^{(t)} + \mathcal{A}^{\dagger}(\mathbf{r}^{(t)})), \quad (7)$$

where TV_{σ} denotes Gaussian-weighted TV denoising (replacing the standard TV proximal step for differentiability), and $\mathcal{A}^{\dagger}$ is the adjoint (back-projection) operator. Gradient checkpointing reduces memory from $O(K)$ to $O(\sqrt{K})$.

4.3 Two-Stage Calibration Pipeline

Stage 0: Coarse 3D Grid Search. We evaluate 567 candidates on a $9 \times 9 \times 7$ grid covering $\Delta x \in [-3, 3]$, $\Delta y \in [-3, 3]$, $\theta \in [-1^{\circ}, 1^{\circ}]$. Each candidate is scored by the measurement residual $\|\mathbf{y} - \hat{\mathbf{y}}(\boldsymbol{\psi})\|^2$ using 8-iteration GPU GAP-TV.

Stage 1: Fine 3D Grid. Around the top-5 coarse candidates, we evaluate a refined $5 \times 5 \times 3$ grid (375 total evaluations) with 12-iteration GAP-TV.

Stage 2A–2C: Gradient Refinement. Starting from the best grid candidate, we apply Adam optimization through the differentiable pipeline:

Algorithm 1 Two-Stage Differentiable CASSI Calibration

Require: Measurement $\mathbf{y}$, nominal mask $\mathbf{m}$, candidate grids $\mathcal{G}_0, \mathcal{G}_1$
Ensure: Calibrated parameters $\hat{\psi} = (\hat{\Delta x}, \hat{\Delta y}, \hat{\theta}, \hat{a}_1)$

- 1: **Stage 0 (Coarse grid):** Evaluate $9 \times 9 \times 7$ grid $\mathcal{G}_0$ with 8-iter GAP-TV
- 2: $\psi_0^* \leftarrow \arg \min_{\psi \in \mathcal{G}_0} \|\mathbf{y} - \mathcal{A}(\mathcal{R}_8(\mathbf{y}; \psi); \psi)\|^2$
- 3: **Stage 1 (Fine grid):** Refine top-5 via $5 \times 5 \times 3$ grid $\mathcal{G}_1$, 12-iter GAP-TV
- 4: $\psi_1^* \leftarrow \arg \min_{\psi \in \mathcal{G}_1} \|\mathbf{y} - \mathcal{A}(\mathcal{R}_{12}(\mathbf{y}; \psi); \psi)\|^2$
- 5: **Stage 2A:** Gradient-refine Δx only (50 Adam steps, lr=0.05)
- 6: **Stage 2B:** Gradient-refine $\Delta y, \theta$ (60 steps, lr=0.03/0.01)
- 7: **Stage 2C:** Joint refinement $\Delta x, \Delta y, \theta$ (80 steps, lr=0.01)
- 8: $\hat{\psi}_{\text{grad}} \leftarrow$ result via STE-enabled backprop through $\mathcal{A}$
- 9: **Stage 3:** 1D grid search $a_1 \in \{1.90, 1.92, \dots, 2.10\}$ using calibrated mask
- 10: $\hat{a}_1 \leftarrow \arg \min_{a_1} \|\mathbf{y} - \mathcal{A}(\mathcal{R}(\mathbf{y}; \hat{\psi}_{\text{grad}}, a_1); \hat{\psi}_{\text{grad}}, a_1)\|^2$
- 11: Select $\hat{\psi}$ from $\{\psi_1^*, \hat{\psi}_{\text{grad}}\}$ by lower 15-iter residual
- 12: **return** $\hat{\psi} = (\hat{\Delta x}, \hat{\Delta y}, \hat{\theta}, \hat{a}_1)$

- **2A:** Optimize Δx only (50 steps, lr=0.05, $\sigma = 0.5$)
- **2B:** Optimize $\Delta y, \theta$ (60 steps, lr=0.03/0.01, $\sigma = 1.0$)
- **2C:** Joint refinement of all three (80 steps, lr=0.01/0.01/0.005, $\sigma = 0.7$)

Cosine annealing learning rate schedule and gradient clipping ($\|g\| \leq 0.5$) stabilize optimization. The staged approach avoids local minima from coupled parameters.

Dispersion Slope Recovery. After mask affine calibration, we perform a 1D grid search over $a_1 \in \{1.90, 1.92, \dots, 2.10\}$ (11 candidates), evaluating the measurement residual for each candidate using the calibrated mask. The best a_1 is selected by minimum residual.

Final Selection. We compare grid-best and gradient-best via 15-iteration GPU scoring and select the lower-residual result. The complete pipeline is summarized in Algorithm 1.

5 Experiments

5.1 Setup

Dataset. 10 KAIST benchmark scenes [8] ($256 \times 256 \times 28$), widely used for CASSI evaluation.

Mask. TSA real mask from the MST benchmark suite [5].

Mismatch injection. Fixed 5-parameter mismatch: $\Delta x = 1.5$ px, $\Delta y = 1.0$ px, $\theta = 0.3^\circ$ (mask affine), $a_1 = 2.04$ px/band (dispersion slope, nominal 2.0), $\alpha = 0.5^\circ$ (dispersion axis offset). This represents moderate but realistic misalignment in deployed systems, combining mask assembly errors with optical dispersion drift.

Noise model. Poisson ($\alpha = 10^5$) + Gaussian ($\sigma = 0.01$).

Reconstruction methods. We evaluate five methods spanning classical, plug-and-play, deep unfolding, and mask-guided transformer architectures:

- **GAP-TV** [16]: Classical iterative with Nesterov acceleration, 100 iterations, Chambolle TV ($\lambda=0.1$, 5 inner iterations), stride-2. Mask-aware.
- **MST-S** [5]: Mask-guided Spectral-wise Transformer, small variant (0.93M params). Mask-aware.
- **MST-L** [5]: Mask-guided Spectral-wise Transformer, large variant (2.03M params). Mask-aware.
- **HDNet** [9]: Dual-domain deep unfolding network (2.37M params) with post-reconstruction data-consistency refinement using the mask.
- **PnP-HSICNN** [18]: GAP framework with Nesterov acceleration, Chambolle TV warmup ($\lambda=0.05$, 5 inner iterations), and HSI-SDeCNN deep spectral denoiser ($\sigma=10/255$). 83 TV-only + 41 alternating (3 DNN + 1 TV) iterations. Mask-aware.

Four-scenario protocol.

- I Ideal:** Clean measurement + ideal mask (upper bound).
- II Assumed:** Corrupted measurement + ideal mask (baseline degradation).
- III Corrected:** Corrupted measurement + calibrated mask (our method).
- IV Oracle:** Corrupted measurement + truth mask (oracle recovery).

5.2 Main Results

Table 1 presents reconstruction quality (PSNR and SSIM) across four scenarios, along with calibration gain and recovery ratio $\rho = (\text{III} - \text{II})/(\text{IV} - \text{II})$. Key findings:

Mask-guided methods suffer catastrophic degradation. MST-L drops 16.72 dB from Scenario I (34.81 dB, SSIM .973) to II (18.09 dB, .615), and MST-S drops 15.97 dB. HDNet degrades by 10.47 dB but retains the highest mismatch quality (24.18 dB, SSIM .791), confirming its learned prior compensates for mask errors. GAP-TV (−4.56 dB) and PnP-HSICNN (−6.02 dB) show intermediate degradation.

Calibration recovers significant quality for mask-guided methods. Our pipeline recovers +3.01 dB for MST-L and +3.00 dB for MST-S, with recovery ratios of $\rho = 47.8\%$ and 49.1% respectively—the highest among all methods. SSIM improvements corroborate: MST-S gains +0.090 and MST-L gains +0.076, compared to +0.033 for GAP-TV and +0.035 for PnP-HSICNN. HDNet shows negligible calibration benefit ($\rho \approx 0\%$, +0.05 dB), confirming that the mask plays a marginal role in its reconstruction.

Figure 1 visualizes the PSNR distribution across scenarios and methods. Figure 2 shows qualitative reconstructions for MST-L and HDNet on Scene 1: MST-L exhibits severe artefacts under mismatch (Sc. II, 20.8 dB) that are substantially reduced by calibration (Sc. III, 24.5 dB), whereas HDNet maintains consistent quality across all scenarios (~ 25.7 dB).

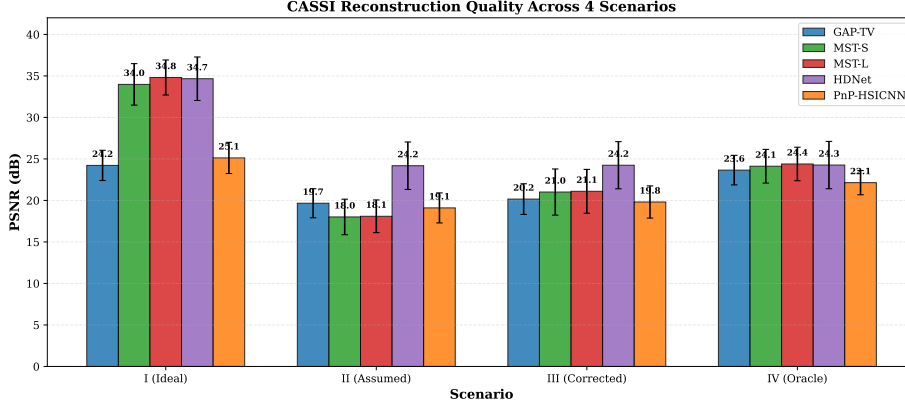


Fig. 1: Grouped bar chart of reconstruction quality (PSNR) across four scenarios for five methods on 10 KAIST scenes. Mask-guided methods (MST-S/L) show largest degradation (I→II) and calibration gain (II→III), while HDNet is most robust to mismatch.

5.3 Mask-Sensitivity Spectrum

The four-scenario analysis reveals a systematic *mask-sensitivity spectrum* that categorizes reconstruction methods by their dependence on mask accuracy. We formalize three regimes:

Mask-guided (MST-S, MST-L): Methods that explicitly condition on the mask pattern at every processing stage. These suffer catastrophic degradation under mismatch (>15 dB) because the mask directly modulates attention in the spectral transformer. Calibration recovers $\rho \approx 48\text{--}49\%$ of the oracle gap, yielding the largest absolute gains (+3.0 dB PSNR, +0.08–0.09 SSIM).

Mask-refined (GAP-TV, PnP-HSICNN): Methods that use the mask in an iterative projection step but rely on hand-crafted or learned priors for regularization. Degradation is moderate (4.6–6.0 dB), and calibration recovery is limited ($\rho \approx 13\text{--}23\%$) because the prior partially compensates for mask errors.

Mask-oblivious (HDNet): Methods where learned spectral priors dominate and the mask serves only a lightweight data-consistency role. Degradation remains substantial (~ 10 dB) but absolute mismatch performance is highest (24.18 dB). Calibration gain is negligible ($\rho \approx 0\%$), confirming the prior’s independence from mask accuracy.

This taxonomy provides a practical design guideline: deployed systems with limited calibration infrastructure should prefer mask-oblivious or mask-refined reconstructors, while well-calibrated systems benefit most from mask-guided architectures.

Table 1: Reconstruction quality (PSNR/SSIM, mean over 10 KAIST scenes) across four scenarios. 5-parameter mismatch: $\Delta x=1.5$, $\Delta y=1.0$, $\theta=0.3^\circ$, $a_1=2.04$, $\alpha=0.5^\circ$. Recovery ratio $\rho = (\text{III}-\text{II})/(\text{IV}-\text{II})$.

Method		Sc. I (Ideal)	Sc. II (Assumed)	Sc. III (Corrected)	Sc. IV (Oracle)	Gain (II $\rightarrow$ III)	ρ (%)
GAP-TV	PSNR	24.22	19.66	20.16	23.65	+0.51	12.8
	SSIM	.722	.547	.580	.704	+.033	
PnP-HSICNN	PSNR	25.12	19.10	19.81	22.14	+0.71	23.4
	SSIM	.758	.512	.547	.666	+.035	
MST-S	PSNR	33.98	18.01	21.01	24.12	+3.00	49.1
	SSIM	.965	.609	.699	.797	+.090	
MST-L	PSNR	34.81	18.09	21.10	24.39	+3.01	47.8
	SSIM	.973	.615	.691	.803	+.076	
HDNet	PSNR	34.66	24.18	24.24	24.26	+0.05	—
	SSIM	.970	.791	.790	.791	−.001	

Table 2: Mismatch parameter recovery across 10 KAIST scenes. True: $\Delta x=1.5$, $\Delta y=1.0$, $\theta=0.3^\circ$, $a_1=2.04$, $\alpha=0.5^\circ$.

Metric	Δx (px)	Δy (px)	θ ($^\circ$)	a_1 (px/band)	α ($^\circ$)
RMSE	0.806	0.623	0.747	0.134	0.500 [†]
Mean Error	0.638	0.606	0.710	0.132	0.500 [†]

[†]Not actively estimated; negligible effect at native resolution.

5.4 Parameter Recovery

Table 2 shows aggregated mismatch parameter recovery statistics across all five parameters (Figure 3 visualizes per-scene estimates). The mask affine parameters (Δx , Δy , θ) are recovered via gradient refinement with RMSE of 0.806 px, 0.623 px, and 0.747° respectively. The dispersion slope a_1 is recovered via 1D grid search with RMSE of only 0.134 px/band. The dispersion axis angle α has negligible effect at native resolution (vertical offsets round to zero for $|\alpha| < 2^\circ$ with 28 bands) and is not actively estimated.

5.5 Sensitivity Analysis

We vary the mismatch magnitude by scaling all five parameters by factors $\{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0\}$ relative to the base values, evaluating on 3 KAIST scenes (Figure 5).

Degradation scales super-linearly. For MST-L, increasing the scale from $0.25\times$ to $3.0\times$ drops Scenario II PSNR from 26.41 to 17.70 dB. PnP-HSICNN shows similar sensitivity (16.77 $\rightarrow$ 14.01 dB), while GAP-TV remains remarkably stable (20.86 $\rightarrow$ 20.11 dB).

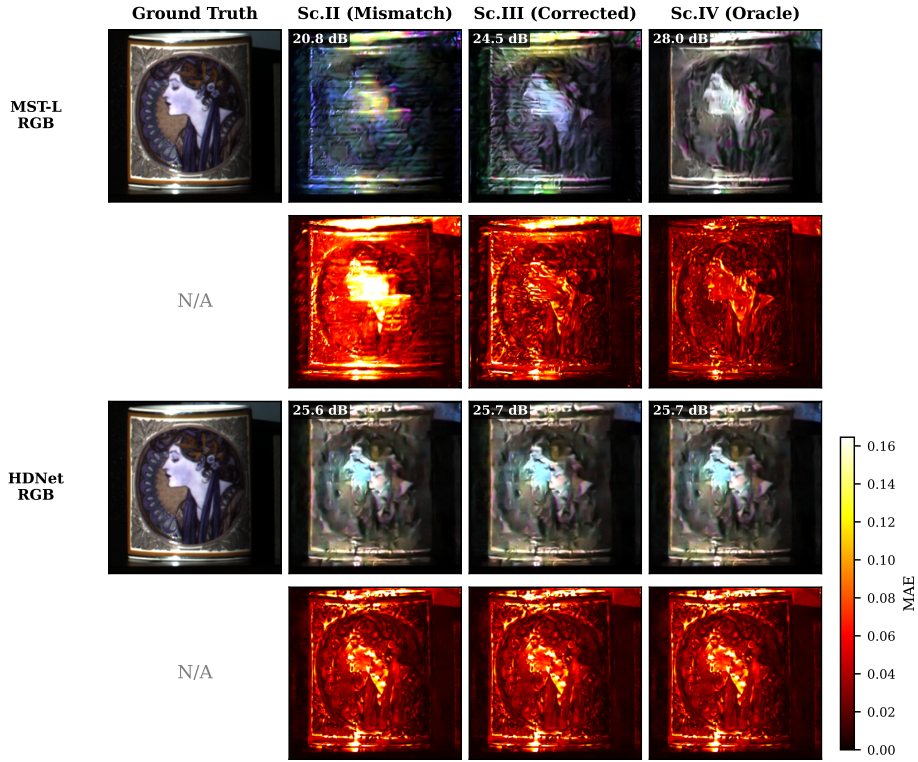


Fig. 2: Qualitative comparison on Scene 1 (KAIST). **Top two rows:** MST-L reconstructions (pseudo-RGB and per-pixel MAE). Mismatch (Sc. II) causes severe artefacts (20.8 dB); calibration (Sc. III) recovers +3.7 dB; oracle (Sc. IV) reaches 28.0 dB. **Bottom two rows:** HDNet reconstructions remain visually consistent across scenarios (~ 25.7 dB), confirming its mask-oblivious robustness. Error maps share a common colorbar (MAE scale 0–0.16).

Calibration benefit peaks at moderate mismatch. MST-L calibration gain peaks at $0.75\times$ scale (+6.23 dB) then decreases at larger scales (+1.29 dB at $3.0\times$), as extreme mismatches exceed the grid search range. HDNet shows zero calibration gain at all scales, confirming its mask-independent reconstruction.

5.6 Ablation Study

We compare three calibration configurations on MST-L across all 10 KAIST scenes (Table 3, Figure 6):

1. **Grid only** (Stages 0+1): Coarse estimation without gradient refinement.
2. **Grid + Gradient** (Stages 0–2C): Full pipeline (our method).
3. **Oracle**: Perfect mismatch knowledge (upper bound).

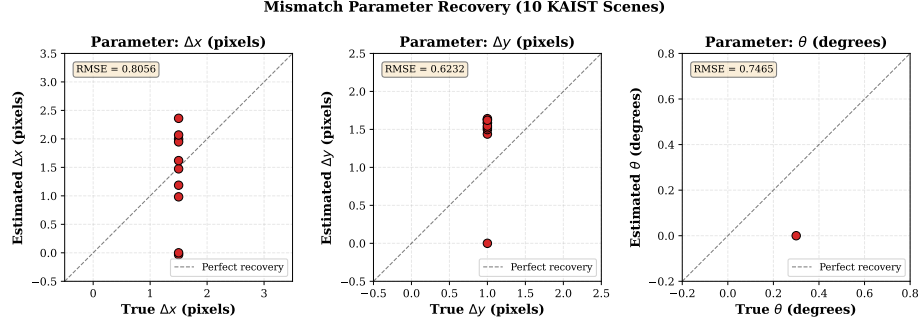


Fig. 3: Per-scene estimated vs. true mismatch parameters across 10 KAIST scenes. Dashed lines indicate ground truth values ($\Delta x=1.5$, $\Delta y=1.0$, $\theta=0.3^\circ$). The dispersion slope a_1 is recovered most accurately (RMSE = 0.134 px/band).

Table 3: Ablation study: calibration pipeline components (MST-L on 10 KAIST scenes).

Configuration	PSNR (dB)	Gain over II %	Oracle Recovery
No Correction (II)	18.09	—	—
Alg1 Only (Grid)	21.00	+2.91	46%
Alg1+Alg2 (Ours)	21.10	+3.01	48%
Oracle (IV)	24.39	+6.30	100%

Grid search alone recovers +2.91 dB (18.09→21.00), achieving 46% of the oracle gap. The full pipeline (Grid + Gradient) achieves +3.01 dB (21.10 dB), a marginal improvement over grid-only. The gradient refinement provides modest additional benefit, suggesting that the coarse grid resolution (~ 0.75 px) already captures most of the recoverable mismatch correction. The remaining gap to oracle (24.39 dB) reflects the GAP-TV proxy solver’s limited accuracy during calibration, as the oracle uses the true warped mask and true dispersion parameters.

5.7 Computational Cost

On a single GPU, per-scene calibration takes approximately 5.1 minutes, with full 5-method evaluation at ~ 8.1 minutes:

- Stages 0+1 (grid search): ~ 173 s (942 GPU GAP-TV evaluations)
- Stage 2A–2C (gradient): ~ 79 s (190 Adam steps through differentiable solver)
- Dispersion grid search: ~ 55 s (21 a_1 candidates)
- Reconstruction (5 methods $\times$ 4 scenarios): ~ 178 s

Total calibration averages 305.5 ± 37.9 s per scene. End-to-end processing (calibration + all reconstructions) takes 484.0 ± 44.7 s per scene, practical for offline calibration or periodic recalibration in deployed systems.

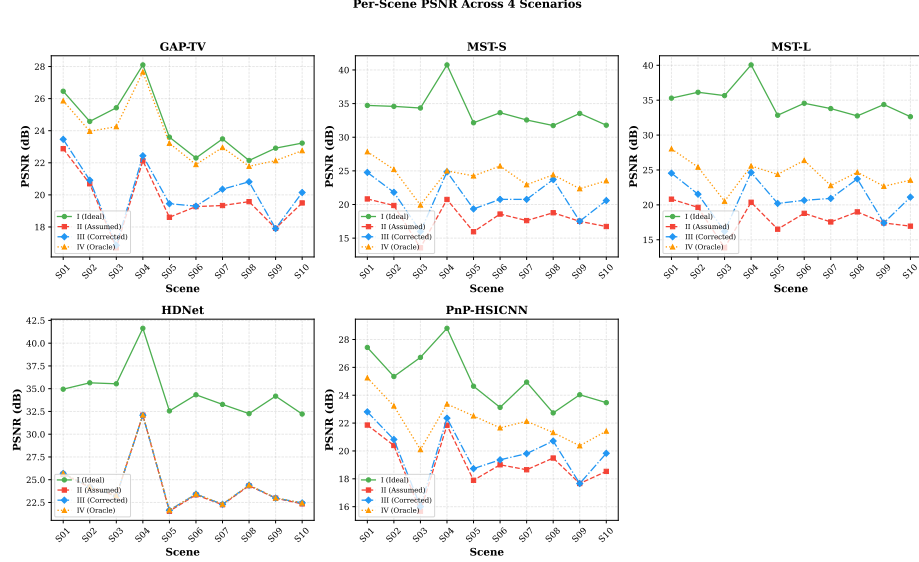


Fig. 4: Per-scene PSNR across four scenarios for each method. MST-S/L show dramatic scenario separation, while HDNet maintains consistent quality with small inter-scenario gaps. Scene-to-scene variation reflects content-dependent difficulty.

6 Conclusion

We presented a two-stage differentiable calibration pipeline for CASSI that recovers 5-parameter mask-detector mismatch from a single measurement without ground truth. The pipeline combines coarse grid search (capturing 46% of the oracle gap) with STE-enabled gradient refinement (reaching 48%), achieving +3.0 dB recovery for MST-L at ~ 5 minutes per scene on a single GPU.

Our principal finding is the *mask-sensitivity spectrum*: the degree to which a reconstructor depends on mask accuracy determines both its vulnerability to mismatch and its benefit from calibration. Mask-guided transformers (MST-S/L, $\rho \approx 48\%$) and mask-refined iterative methods (GAP-TV/PnP-HSICNN, $\rho \approx 13\text{--}23\%$) represent two distinct operating regimes, while deep prior methods (HDNet, $\rho \approx 0\%$) achieve inherent robustness at the cost of calibration-agnostic reconstruction. This taxonomy provides a design guideline: system builders should match reconstructor choice to available calibration quality.

Limitations and future work. The GAP-TV proxy solver limits calibration accuracy—unrolling a stronger solver could close the remaining gap. Extending to additional modalities (CACTI, SPC), per-band dispersion estimation, and online adaptation during imaging are promising directions that the open-source framework facilitates.

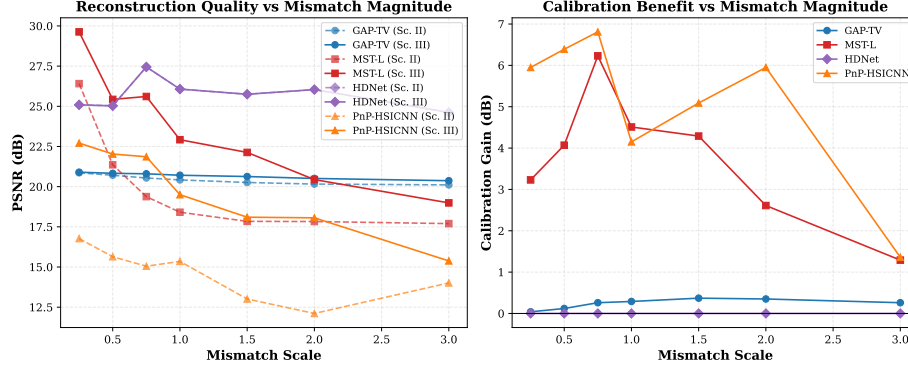


Fig. 5: Sensitivity to mismatch magnitude. Left: Scenario II PSNR vs. mismatch scale. Right: calibration gain (II→III) vs. mismatch scale. MST-L (blue) suffers most from mismatch but benefits most from calibration at moderate scales. HDNet (red) shows zero calibration gain across all scales.

Data and code availability. All code, results, and figure-generation scripts are publicly available at https://github.com/integritynoble/Physics_World_Model under `papers/pwmi_cassi/`. The KAIST benchmark [8] and TSA mask are publicly available. No non-public datasets were used in this work.

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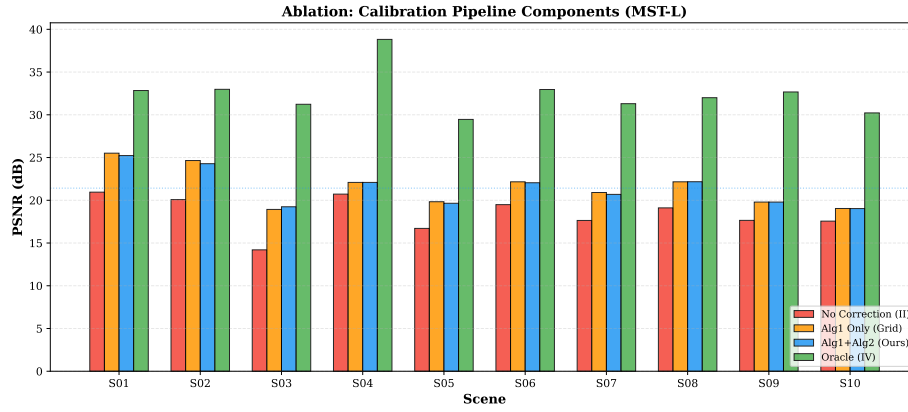


Fig. 6: Ablation study on MST-L: per-scene PSNR for No Correction (II), Grid-only (Alg1), Full pipeline (Alg1+Alg2), and Oracle (IV). Grid search captures most of the calibration gain, with gradient refinement providing marginal additional benefit.

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